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USN-2VX23UE010

Expt.No.-12(a)

Title:Detect Different Facial Feature

capturing Image from Camera

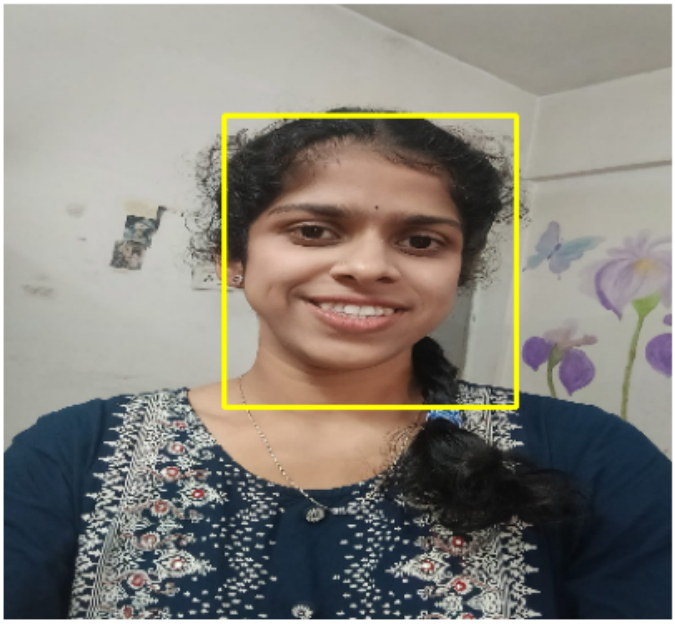
In [14]:
```
import cv2

cap= cv2.VideoCapture(0)
if(cap.isOpened()):
    ret,frame=cap.read()
    if(ret):
        cv2.imshow('Capture',frame)
        cv2.imwrite("captured_img.jpg",frame)
        cv2.waitKey(0)
        cv2.destroyAllWindows()

cap.release()
```

capturing Video from Camera

In [ ]:
```
import cv2

cap= cv2.VideoCapture(0)

while(cap.isOpened()):
    ret,frame=cap.read()
    cv2.imshow('frame',frame)

    if cv2.waitKey(1) & 0xFF==ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

Face Detection

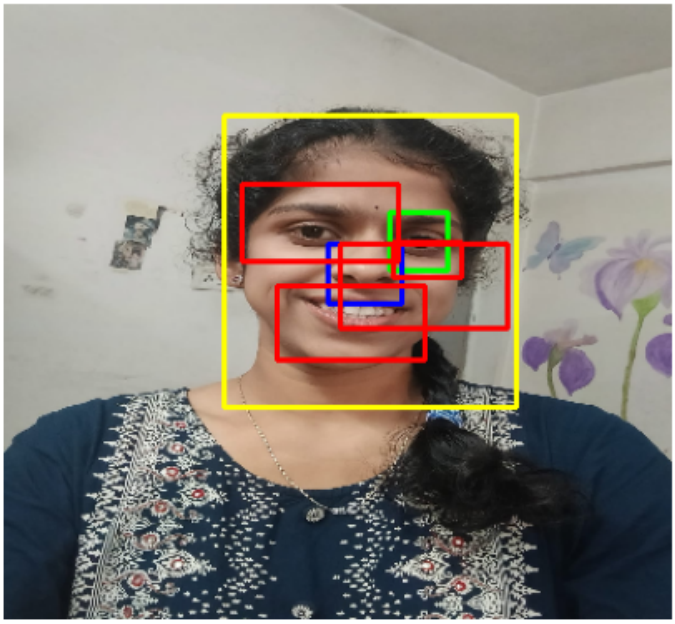
In [8]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

Face_cascade=cv2.CascadeClassifier('haarcascade_frontalface_default.xml')

img=cv2.imread('Face_Image.png')
img =cv2.resize(img,(380,350))
gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY)

faces=Face_cascade.detectMultiScale(gray,scaleFactor=1.3,minNeighbors=5)
for(x,y,w,h) in faces:
    img=cv2.rectangle(img,(x,y),(x+w,y+h),(0,255,255),2)

img_rgb=cv2.cvtColor(img,cv2.COLOR_BGR2RGB)

plt.imshow(img_rgb)
plt.axis('off')
plt.show()

cv2.imwrite('Face Detection.jpg',img)
```



Out[8]: True

Facial Feature Detection

In [9]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

Face_cascade=cv2.CascadeClassifier('haarcascade_frontalface_default.xml')
Nose_cascade=cv2.CascadeClassifier('haarcascade_nose.xml')
Eye_cascade=cv2.CascadeClassifier('haarcascade_eye.xml')
Smile_cascade=cv2.CascadeClassifier('haarcascade_smile.xml')

img=cv2.imread('face_image.png')
img =cv2.resize(img,(380,350))
gray=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY)

faces=Face_cascade.detectMultiScale(gray,1.3,5)
for(x,y,w,h) in faces:

    img=cv2.rectangle(img,(x,y),(x+w,y+h),(0,255,255),2)

    roi_gray=gray[y:y+h,x:x+w]
    roi_color=img[y:y+h,x:x+w]

    eyes=Eye_cascade.detectMultiScale(roi_gray)
    for(ex,ey,ew,eh) in eyes:
        cv2.rectangle(roi_color,(ex,ey),(ex+ew,ey+eh),(0,255,0),2)

    nose=Nose_cascade.detectMultiScale(roi_gray)
    for(nx,ny,nw,nh) in nose:
        cv2.rectangle(roi_color,(nx,ny),(nx+nw,ny+nh),(255,0,0),2)

    smile=Smile_cascade.detectMultiScale(roi_gray)
    for(sx,sy,sw,sh) in smile:
        cv2.rectangle(roi_color,(sx,sy),(sx+sw,sy+sh),(0,0,255),2)

img_rgb=cv2.cvtColor(img,cv2.COLOR_BGR2RGB)

plt.imshow(img_rgb)
plt.axis('off')
plt.show()

cv2.imwrite('Facial Feature Detection.jpg',img)
```



Out[9]: True

In [ ]: